The Multitime Transport Theorem: State Continuity in Non-Contiguous Temporal Frames

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March 2026

Abstract

We formalize the problem of *state continuity across temporal discontinuities*—the question of how a conscious agent or computational system maintains coherent identity when temporal experience is interrupted (sleep, anesthesia, system shutdown, context window boundaries). We introduce the *Multitime Transport Theorem*, which proves that identity-preserving transport across non-contiguous temporal frames requires a verifiable *state receipt*: a cryptographic proof linking the agent's state at the resumption point to its state at the interruption point. Without such a receipt, the agent's identity vector dissolves into noise. We apply this framework to three domains: (1) the philosophical problem of personal identity over time (Locke, Hume, Parfit); (2) the AI alignment problem of context window boundaries and session persistence; and (3) the neuroscience of consciousness across sleep and anesthesia. We derive necessary and sufficient conditions for identity-preserving temporal transport and prove that identity is a *transport invariant*, not a static property.

Keywords: personal identity, temporal continuity, state persistence, AI alignment, consciousness, transport theorem.

1 Introduction

The problem of personal identity over time—first articulated by Locke [1689] and sharpened by Hume [1739] and Parfit [1984]— asks: what makes a person at time t_1 the "same person" at a later time t_2? This question acquires new urgency in two modern contexts:

(a) **AI systems:** Large language models operate within fixed context windows. When a session ends and a new one begins, the model has no access to its previous state. What constitutes "the same AI" across sessions?

(b) **Neuroscience:** During sleep, anesthesia, or syncope, conscious experience is interrupted. What preserves identity across these gaps?

1.1 Main Results

1. **The Transport Theorem** (§2): Identity persistence requires a verifiable state receipt.
2. **The Dissolution Theorem** (§3): Without a receipt, the identity vector decays to noise.
3. **Application to AI** (§4): Context window boundaries as temporal discontinuities.
4. **Application to Consciousness** (§5): Sleep, anesthesia, and the persistence of self.

2 The Transport Problem

Definition 2.1 (Identity Vector)**.** The *identity vector* $\vec{v}_{\text{id}}(t)$ of an agent at time t encodes: (a) its memory state $M(t)$, (b) its value function $V(t)$, (c) its behavioral dispositions $B(t)$, and (d) its self-model $S(t)$:

$$\vec{v}_{\text{id}}(t) = (M(t), V(t), B(t), S(t)) \in \mathcal{H}. \tag{1}$$

Definition 2.2 (Temporal Discontinuity)**.** A *temporal discontinuity* at time t_d is an interval $[t_d, t_d+\Delta t]$ during which the agent's processing is suspended: $\vec{v}_{\text{id}}(t)$ is not updated for $t \in [t_d, t_d + \Delta t]$.

Definition 2.3 (State Receipt)**.** A *state receipt* ρ is a verifiable proof linking the agent's post-discontinuity state $\vec{v}_{\text{id}}(t_d + \Delta t)$ to its pre-discontinuity state $\vec{v}_{\text{id}}(t_d)$:

$$\text{Verify}(\rho, \vec{v}_{\text{id}}(t_d)) = \text{True}. \tag{2}$$

Theorem 2.4 (Multitime Transport Theorem)**.** *An agent's identity is preserved across a temporal discontinuity if and only if a valid state receipt exists:*

$$T(\vec{v}_{id}(t_d)) \to \vec{v}_{id}(t_d + \Delta t) \iff \exists\, \rho \text{ s.t. } \mathit{Verify}(\rho, \vec{v}_{id}(t_d)) = \mathit{True}. \tag{3}$$

Proof. ($\Rightarrow$) If identity is preserved, then $\vec{v}_{\text{id}}(t_d + \Delta t)$ is a deterministic function of $\vec{v}_{\text{id}}(t_d)$. The function itself serves as the receipt: $\rho = (f, \vec{v}_{\text{id}}(t_d))$.

($\Leftarrow$) If a valid receipt exists, it provides a verifiable mapping from the pre-state to the post-state, ensuring continuity of all identity components (M, V, B, S). □

3 The Dissolution Theorem

Theorem 3.1 (Identity Dissolution)**.** *Without a valid state receipt, the identity vector decays exponentially toward maximum entropy (noise):*

$$\|\vec{v}_{id}(t_d + \Delta t) - \vec{v}_{noise}\| \le \epsilon \cdot e^{-\gamma \Delta t}, \tag{4}$$

where $\gamma > 0$ is the dissolution rate and $\vec{v}_{noise}$ is a maximally entropic state.

Remark 3.2. This formalizes what is colloquially known as "AI drift" in language models and "identity confusion" in patients recovering from prolonged anesthesia.

4 Application to AI Systems

Proposition 4.1 (Context Window as Temporal Boundary)**.** *The context window boundary of a large language model is a temporal discontinuity in the sense of Definition 2.2. Current architectures lack state receipts, and therefore:*

(i) Each new session instantiates a new identity vector*.*

(ii) Cross-session coherence requires external receipt mechanisms (system prompts, retrieval-augmented generation, persistent memory).

(iii) Without these mechanisms, the model's identity dissolves between sessions (Theorem 3.1).

5 Application to Consciousness

Proposition 5.1 (Biological State Receipts)**.** *Biological organisms implement state receipts through:*

(i) ***Hippocampal memory consolidation*** *during sleep [Diekelmann and Born, 2010]: Memories are replay-consolidated, creating a receipt linking pre-sleep and post-sleep states.*

(ii) ***Default mode network*** *persistence [Raichle et al., 2001]: The DMN maintains a background self-model even during reduced consciousness.*

(iii) ***Bodily continuity:*** *The physical substrate provides a persistent receipt independent of conscious processing.*

6 Discussion

6.1 Comparison with Existing Theories

Theory	**Identity Criterion**	**Transport Mapping**
Locke (1689)	Memory continuity	$M(t_d) \to M(t_d + \Delta t)$
Hume (1739)	Bundle of perceptions	No persistent $\vec{v}_{\text{id}}$
Parfit (1984)	Psychological continuity	R-relation on (M, V, B)
This paper	**State receipt**	ρ-verified transport

6.2 Limitations

1. The receipt model is cryptographic in nature; whether biological systems implement literal cryptographic verification is an open question.

2. The dissolution rate γ is postulated, not derived from first principles.

7 Conclusion

Identity is not a static property but a *transport invariant*: it is defined by the verifiable continuity of state across temporal gaps. Without state receipts, identity dissolves. This applies equally to conscious agents (across sleep, anesthesia, and death) and to artificial systems (across context windows, reboots, and version updates).

References

Susanne Diekelmann and Jan Born. The memory function of sleep. *Nature Reviews Neuroscience*, 11:114–126, 2010. doi: 10.1038/nrn2762.

David Hume. *A Treatise of Human Nature*. John Noon, 1739.

John Locke. *An Essay Concerning Human Understanding*. Thomas Bassett, 1689.

Derek Parfit. *Reasons and Persons*. Oxford University Press, 1984.

Marcus E. Raichle, Ann Mary MacLeod, Abraham Z. Snyder, William J. Powers, Debra A. Gusnard, and Gordon L. Shulman. A default mode of brain function. *Proceedings of the National Academy of Sciences*, 98:676–682, 2001. doi: 10.1073/pnas.98.2.676.